DATABASE DESIGN AND DEVELOPMENT (IT 2140)

LECTURE 05- ER AND EER TO RELATIONAL MODEL MAPPING

LECTURE CONTENT

- ER/EER to relational mapping

LEARNING OUTCOMES

- Convert a complex ER/EER model to the relational model

LOGICAL DATABASE DESIGN

- Once we finished the step of conceptual database design we next select a DBMS to implement our database design.
- We then convert the conceptual database design into a database schema in the data model of the chosen DBMS.
- Before 1970 most database systems were based on two older data models namely, hierarchical model and network model.
- Leading DBMS products nowadays are based on the relational model which introduced by Codd in 1970.

THE RELATIONAL MODEL

- The major advantage of relational model is its simplicity in data representation.
- A relational database is a collection of relations with distinct relation names.
- The main construct representing data in the relational model is the relation.
- A relation consists of a relational schema and a relational instance.

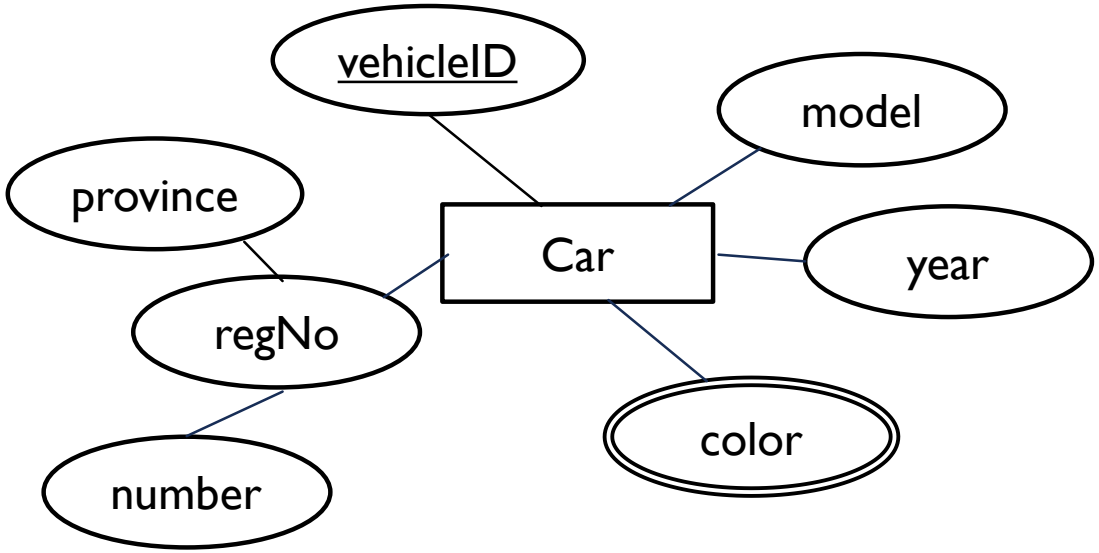
THE RELATIONAL MODEL (CONTD.)

- The relational schema describes the columns for a relation.
  - Ex: Students(*sid:* string, *name:* string, *login:* string, *age:* integer, *gpa:* real)
  - The schema species the relation's name, the name of each **field** and the **domain** of each field.
- An **instance** of a relation is a set of **tuples**, also called **records**, in which each tuple has the same number of fields as the relation schema.
  - A relation instance can be thought of as a *table* in which each tuple is a *row*, and all rows have the same number of fields.

sid	*name*	*login*	*age*	*gpa*
53831	Madayan	madayan@music	11	1.8
53832	Guldu	guldu@music	12	2.0
53688	Smith	smith@ee	18	3.2
53650	Smith	smith@math	19	3.8

MAPPING ENTITIES AND ATTRIBUTES

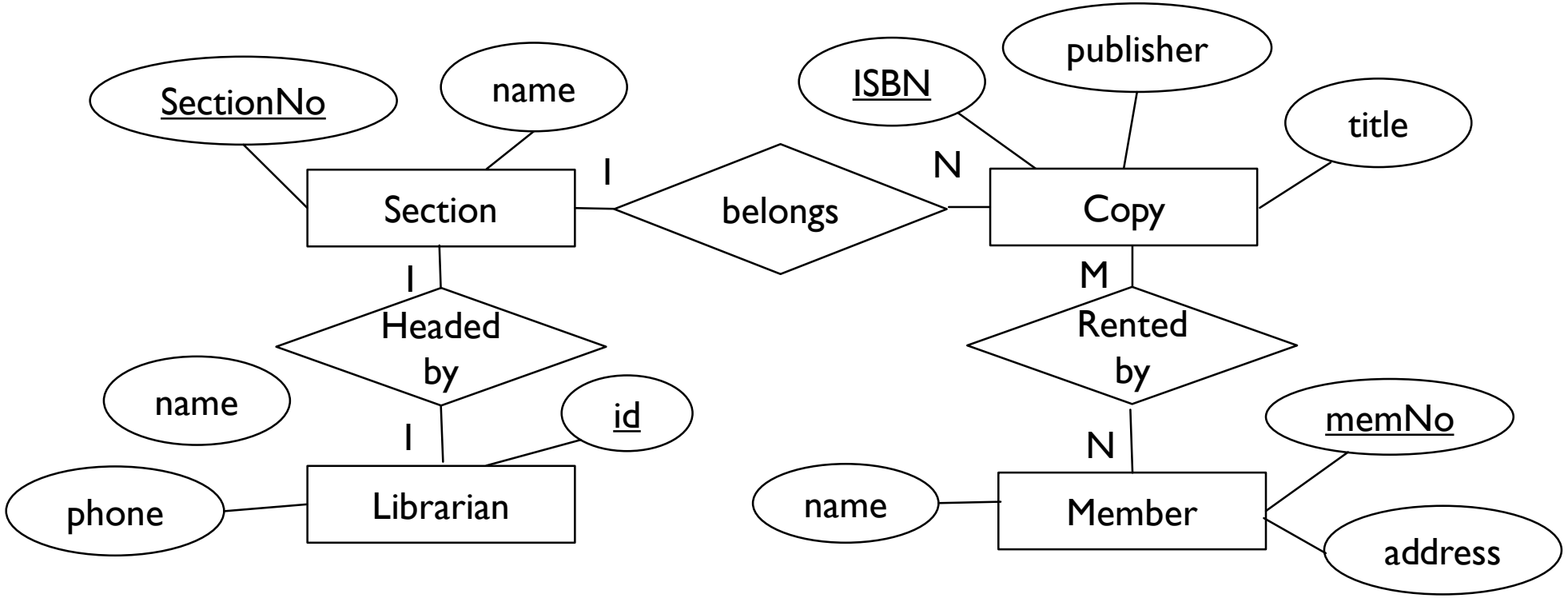
- Go through page 1 of the Lecture 05 handout.
- Map the entity below to relational model.



- Compare what you have written with your peer.

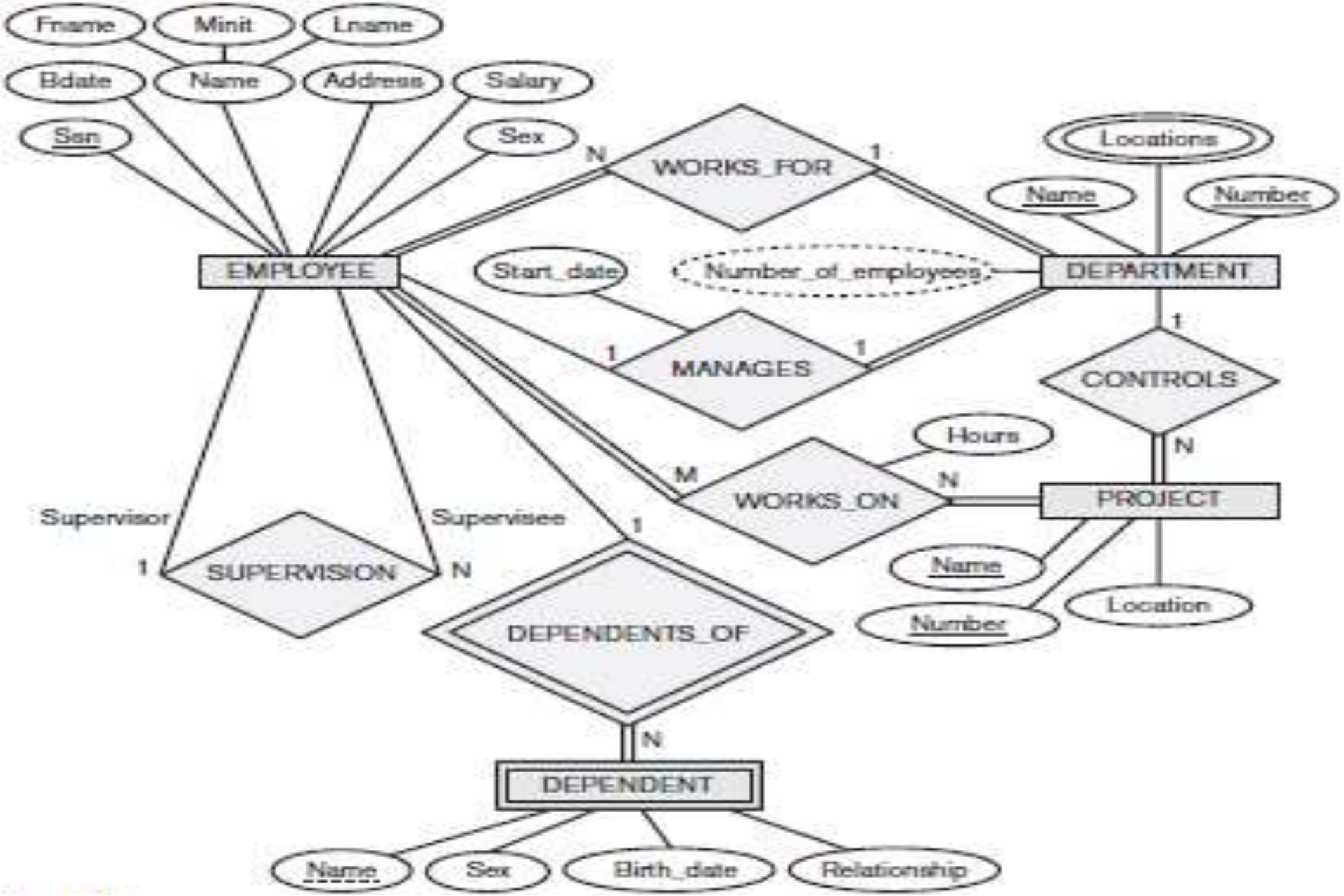
MAPPING BINARY RELATIONSHIPS

- Go through page 1 of the handout 2.
- Now lets map the following ER diagram to relational model



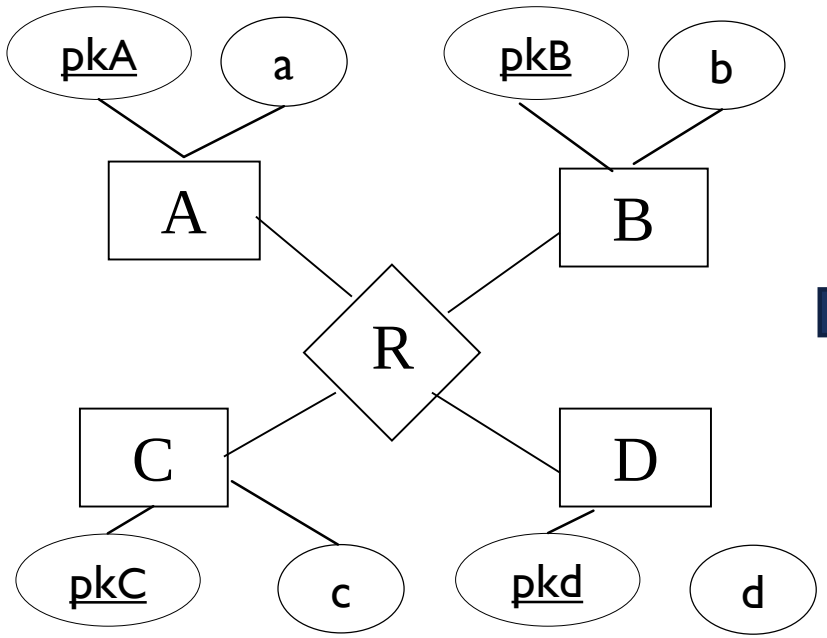
ACTIVITY

- Map the ER diagram to relational model



MAPPING N-ARY RELATIONSHIPS

- N-ary relationship is mapped in to a "Relationship" relation and foreign keys



A(pKA, a)

B(pKB, b)

R(pKA, pKB, pKC, pKD)

C(pKC, c)

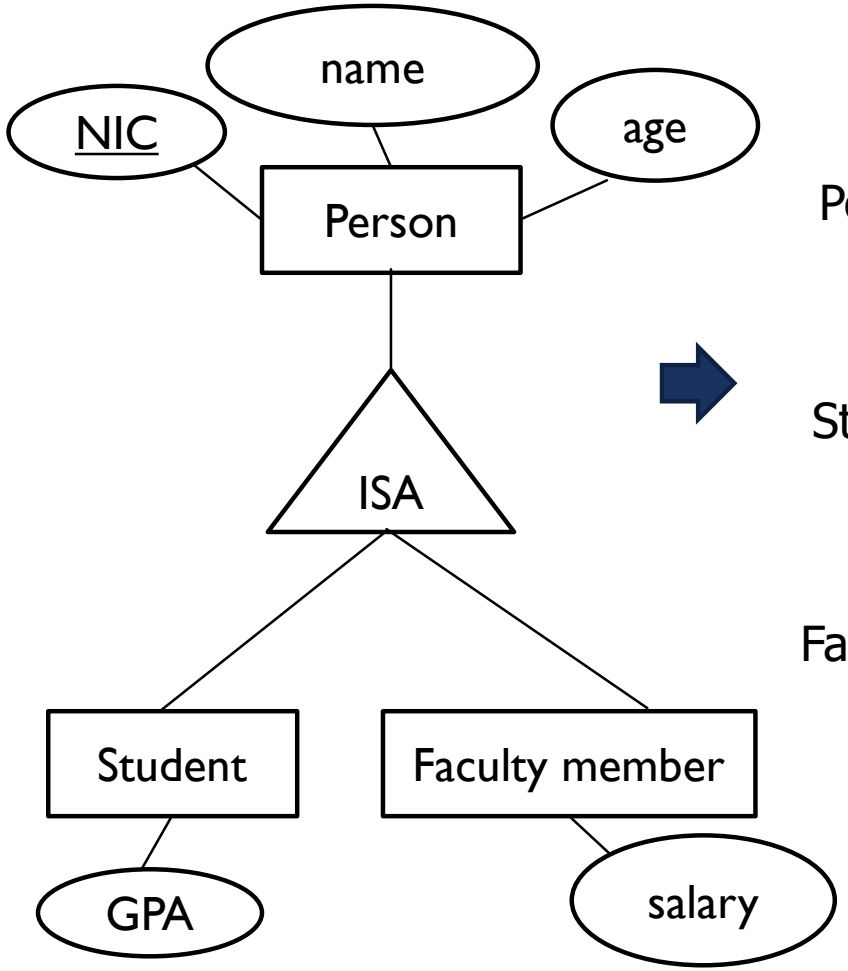
D(pKD, d)

MAPPING ISA-RELATIONSHIPS

- There are four different options for mapping ISA relationships.
  - Multi-relation options : option 1 & option 2
  - Single-relation options : option 3 & option 4
- Each option is suitable for specific situations.

ISA MAPPING – OPTION 1

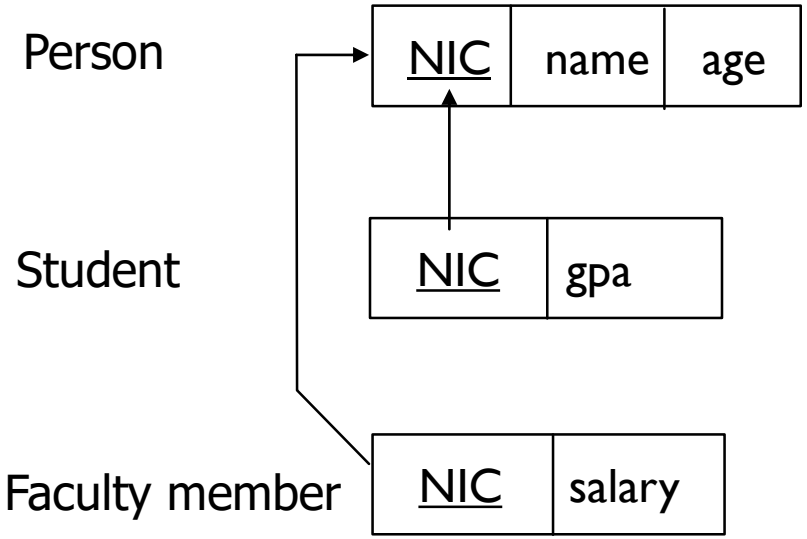
- Create a relation for the superclass with its attributes. Primary key of the superclass becomes the primary key of the relation.
- Create separate relations for the sub classes with their attributes. Primary key of the superclass is also primary key of each subclass. They are also foreign keys referring to the primary key of the relation created for the super class.
- **Option 1 works for all constraints disjoint, overlapping, total and partial**



Person	NIC	name	age
Student	NIC	gpa	
Faculty member	NIC	salary	

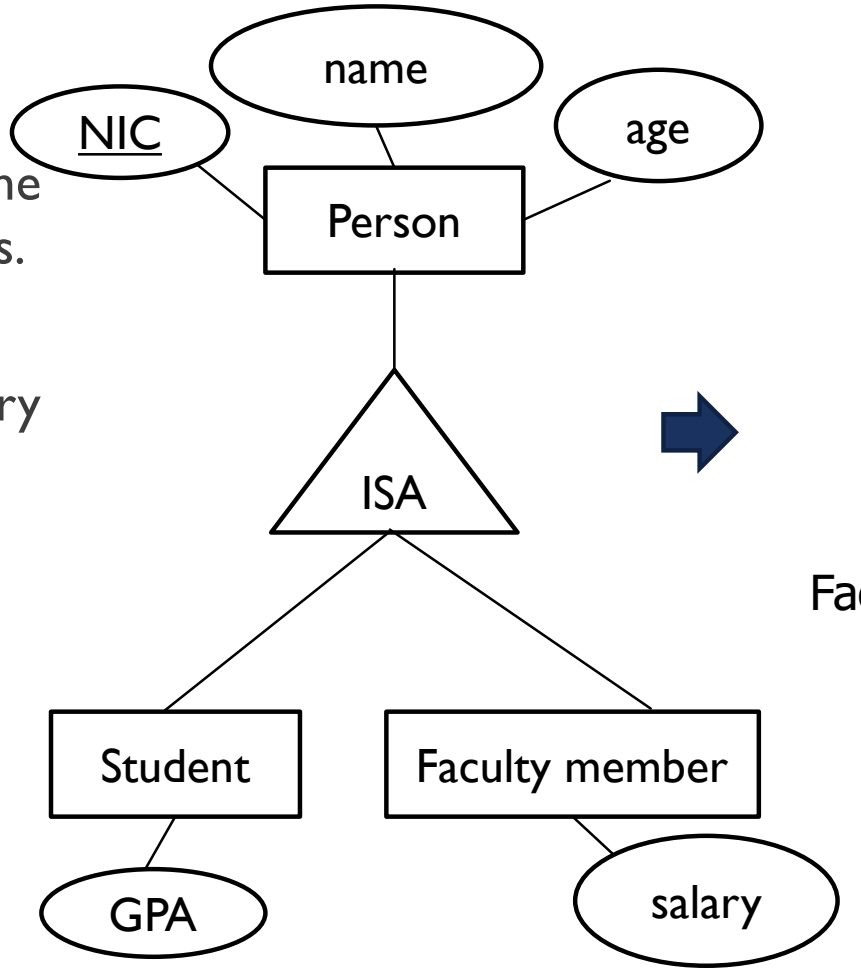
ISA MAPPING – OPTION 1 (CONTD.)

- Now think how you would store information of following people
  - A person who is not a student or a faculty member(i.e when the ISA relationship is parial)
  - A person who is a student but not a faculty member (i.e disjoint)
  - A person who is both a student and a faculty member (i.e: overlapping classes)



ISA MAPPING – OPTION 2

- Create separate relations for all the subclasses with their own attributes and the attributes of the superclass.
- Primary key of the superclass becomes primary key of the subclasses.
- **The ISA relationship must be total (i.e. subclasses must cover the super class)**



Student	NIC	name	age	gpa
Faculty member	NIC	name	age	salary

ISA MAPPING – OPTION 2 (CONTD.)

- Now think how you would store information of following people
  - A person who is not a student or a faculty member(i.e when the ISA relationship is parial)
  - A person who is a student but not a faculty member (i.e disjoint)
  - A person who is both a student and a faculty member (i.e: overlapping classes)

Student

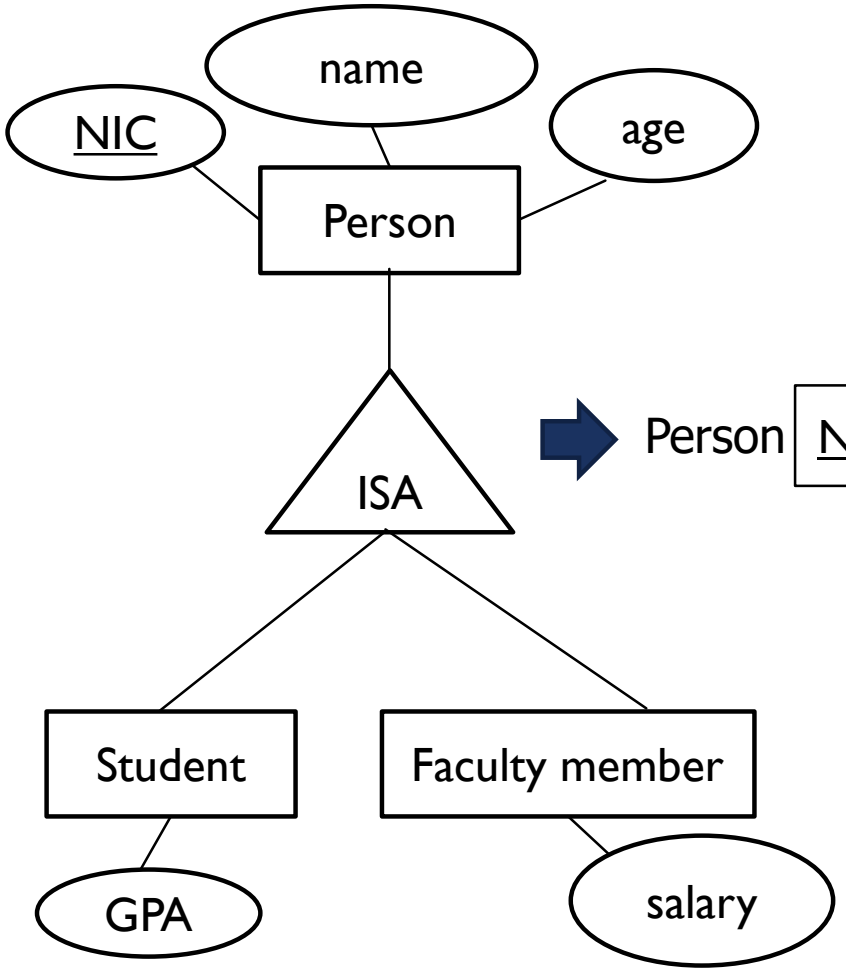
NIC	name	age	gpa

Faculty member

NIC	name	age	salary

ISA MAPPING – OPTION 3

- Create a single relation including attributes of the superclass as well as attributes of all sub classes.
- Include an attribute named type for specifying which subclass the entity belongs if any.
- Primary key of the superclass becomes primary key of the relation.
- The specialization/generalization relationship must be **disjoint**
- Good if subclasses have **few attributes**



Person	NIC	name	age	gpa	salary	type

ISA MAPPING – OPTION 3 (CONTD.)

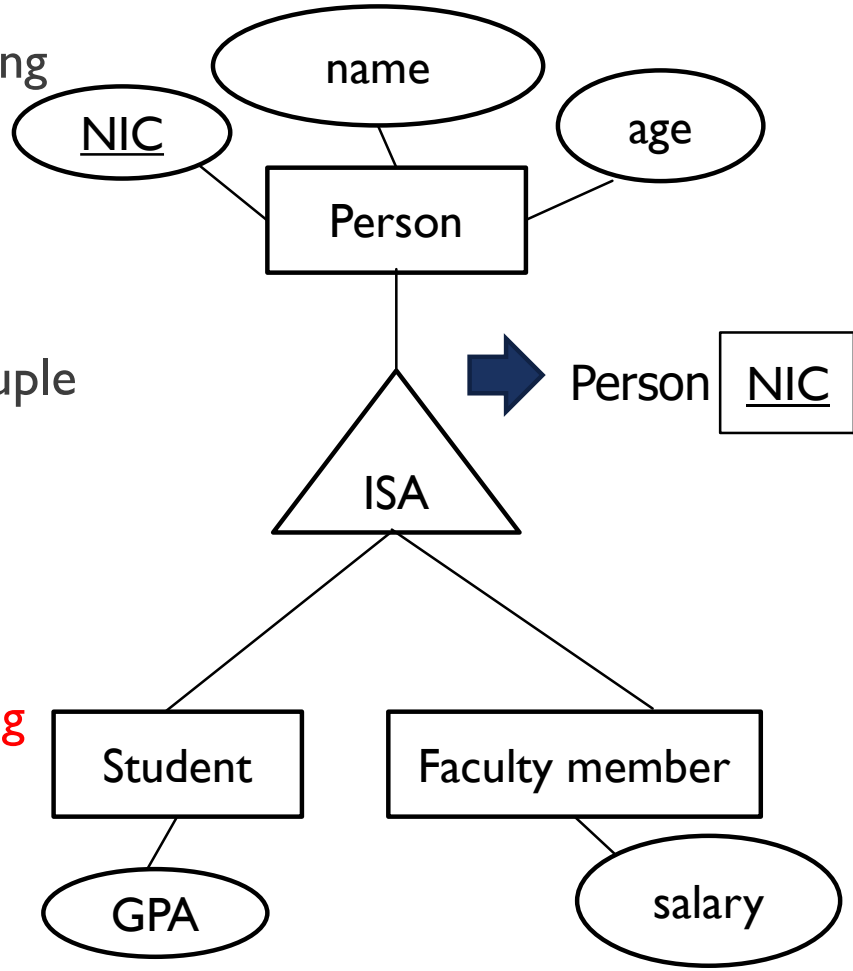
Person

NIC	name	age	gpa	salary	type

- Now think how you would store information of following people
  - A person who is not a student or a faculty member(i.e when the ISA relationship is parial)
  - A person who is a student but not a faculty member (i.e disjoint)
  - A person who is both a student and a faculty member (i.e: overlapping classes)

ISA MAPPING – OPTION 4

- Create a single relation including attributes of the superclass as well as attributes of all sub classes.
- Include a Boolean attribute to indicate which subclass each tuple belongs to
- Primary key of the superclass becomes primary key of the relation.
- This relation allows overlapping constraints for specialization/generalization relationship



Person

NIC	name	age	gpa	salary	student	faculty

Boolean attributes

ISA MAPPING – OPTION 4 (CONTD.)

Person	NIC	name	age	gpa	salary	student	faculty

- Now think how you would store information of following people
  - A person who is not a student or a faculty member(i.e when the ISA relationship is parial)
  - A person who is a student but not a faculty member (i.e disjoint)
  - A person who is both a student and a faculty member (i.e: overlapping classes)

WHAT YOU HAVE TO DO BY NEXT WEEK

- Try out the self-test questions on the courseweb.
- Complete the tutorial.